EXISTENCE AND UNIQUENESS OF LIFTS ALONG DISCRETELY VALUED DIAGRAMS IMPLIES PROPERNESS, BY REDUCING TO THE WEAK CHOW COVER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The separatedness criterion from the same valuative shape, and locality of universal closedness. Same chapter spaces-cohomology.tex, label lemma-check-separated-dvr (invoked at proof line 3710) and Morphisms of Spaces, label spaces-morphisms-lemma-universally-closed-local (invoked at proof line 3713). The first gives separatedness so that properness reduces to universal closedness; the second lets universal closedness be checked etale locally, reducing to a Noetherian affine base.
- The weak form of Chow's lemma. Same chapter spaces-cohomology.tex, label lemma-weak-chow (invoked at proof line 3716). It produces a proper surjective morphism $X' \to X$ with X' a scheme, so that some properties of $X \to Y$ can be tested on X'.
- Inheritance of properness along a proper surjective morphism, and the valuative criterion over a Noetherian base. Morphisms of Spaces, label spaces-morphisms-lemma-image-proper-is-proper (invoked at proof line 3718) and Limits, label limits-lemma-check-universally-closed-Noetherian (invoked at proof line 3720). The first shows the claim that $X' \to \Spec(A)$ is universally closed implies the lemma; the second reduces that claim to lifting dotted arrows for diagrams from the spectrum of a discrete valuation ring.
- The valuative criterion for proper morphisms of schemes. Standard, used at proof lines 3730-3732. Applied to the proper scheme morphism $X' \times_{X, b} \Spec(A) \to \Spec(A)$, it lifts the dotted arrow b to the required arrow a.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0ARJ: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3632–3632. Navigation pointer for \ref{lemma-check-separated-dvr} used inside the counted span (line 3710); the argument that consumes it is retained in the span.

Source reference. Original lemma 089J: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3498–3498. Navigation pointer for \ref{lemma-weak-chow} used inside the counted span (line 3716); the argument that consumes it is retained in the span.

Source reference. Original lemma 05JY: sources/stacks-chow/tags.txt, source lines 7182–7182. Navigation pointer for \ref{limits-lemma-check-universally-closed-Noetherian} used inside the counted span (line 3720); the argument that consumes it is retained in the span.

Source reference. Original lemma 08AJ: sources/stacks-chow/tags.txt, source lines 10534–10534. Navigation pointer for \ref{spaces-morphisms-lemma-image-proper-is-proper} used inside the counted span (line 3718); the argument that consumes it is retained in the span.

Source reference. Original lemma 03IT: sources/stacks-chow/tags.txt, source lines 4615–4615. Navigation pointer for \ref{spaces-morphisms-lemma-universally-closed-local} used inside the counted span (line 3713); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *Y is locally Noetherian,*
(2) *f is of finite type and quasi-separated,*
(3) *for every commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where A is a discrete valuation ring and K its fraction field, there is a unique dotted arrow making the diagram commute.

Then f is proper.

Proof. It suffices to prove f is universally closed because f is separated by Lemma 0ARJ. To do this we may work étale locally on Y (Morphisms of Spaces, Lemma 03IT). Hence we may assume $Y = \mathrm{Spec}(A)$ is a Noetherian affine scheme. Choose $X' \to X$ as in the weak form of Chow's lemma (Lemma 089J). We claim that $X' \to \mathrm{Spec}(A)$ is universally closed. The claim implies the lemma by Morphisms of Spaces, Lemma 08AJ. To prove this, according to Limits, Lemma 05JY it suffices to prove that in every solid commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & X' & \longrightarrow & X \\ \downarrow & \overset{a}{\nearrow} & & \overset{b}{\nearrow} & \downarrow \\ \mathrm{Spec}(A) & & \longrightarrow & & Y \end{array}$$

where A is a dvr with fraction field K we can find the dotted arrow a. By assumption we can find the dotted arrow b. Then the morphism $X' \times_{X,b} \mathrm{Spec}(A) \to \mathrm{Spec}(A)$ is a proper morphism of schemes and by the valuative criterion for morphisms of schemes we can lift b to the desired morphism a. □